MODULE 08 · DOMAIN 1 - SDLC AUTOMATION

Cross-Account and Multi-Region Pipelines

One tooling account releasing into many accounts and Regions: the roles, keys and policies that make it work, and the tools that scale it.

DOMAIN 1 of 6 EXAM WEIGHT 22% MODULE 8 of 34

admin@sthithapragna.com

Every boundary is a door with **two locks**: the caller must be allowed to ask, and the other side must agree to answer. Encrypted artifacts add a **third**: the key.

The locks on a cross-account deployment

LOCK	LIVES IN	MUST ALLOW
Pipeline service role policy	Tooling account	sts:AssumeRole on the target's deploy role
Deploy role trust policy	Target account	The pipeline service role to assume it
Deploy role permissions	Target account	Read the artifact bucket, decrypt with the key
Artifact bucket policy	Tooling account	The target's deploy role to get objects
KMS key policy	Tooling account	The target's deploy role to decrypt

ON THE EXAM

Five places, two accounts. A question usually confirms two of them are correct and hides the fault in a third.

One cross-account deploy, step by step

01

Assume

The pipeline assumes the deploy role in production.

02

Read

The deploy role gets the artifact from the tooling bucket.

03

Decrypt

The customer managed key lets it decrypt.

04

Hand over

It calls CloudFormation, passing an execution role.

05

Create

CloudFormation builds resources as that execution role.

ON THE EXAM

Two roles in the target account: one the pipeline assumes, one CloudFormation works as. They fail differently.

Action role, or execution role?

ACTION ROLE

Assumed by the pipeline

- Set as the action's role ARN
- Reads artifacts, starts the stack operation
- Needs iam:PassRole for the execution role
- Fails as: cannot read or cannot start

CLOUDFORMATION EXECUTION ROLE

Used by CloudFormation

- Set in the action's configuration
- Creates, updates and deletes resources
- Needs permissions for every resource type
- Fails as: stack events show access denied

THE TIEBREAKER

Stack starts, then a resource fails with access denied: the execution role. Nothing starts at all: the action role.

The key policy statement that opens the third lock

```
1  {
2    "Sid": "TargetAccountsDecryptArtifacts",
3    "Effect": "Allow",
4    "Principal": { "AWS": [
5      "arn:aws:iam::222222222222:role/deploy-role",
6      "arn:aws:iam::333333333333:role/deploy-role" ] },
7    "Action": ["kms:Decrypt", "kms:DescribeKey"],
8    "Resource": "*",
9    "Condition": { "StringEquals": {
10     "kms:ViaService": "s3.us-east-1.amazonaws.com" } }
11 }
```

5-6 The deploy roles in staging and production, named explicitly.

10 Only usable through S3 in this Region, not directly.

ON THE EXAM

The AWS managed key for S3 has a policy nobody can edit. That is why cross-account needs a customer managed key.

Artifacts never leave the tooling account

- Every action reads inputs from, and writes outputs to, the pipeline's own artifact store.
- An action in account B writing output needs **put** access to the bucket and **encrypt** on the key.
- Anything that fetches artifacts itself, such as EC2 instances running CodeDeploy, needs the same read and decrypt.
- The key must be in the **same Region** as the pipeline. KMS keys are Regional.

> ON THE EXAM
>
> Build in one target account, deploy in another: both read and write through the tooling bucket and key.

Crossing Regions

- An action can run in another Region by setting its **Region** field.
- The pipeline then needs an **artifact store in that Region**: a bucket and a key there.
- CodePipeline copies artifacts between the Regional stores automatically.
- Replicating the main bucket yourself is not how it works. The stores are declared on the pipeline.

> ON THE EXAM
>
> Cross-Region action fails because no artifact store exists there: add a Regional store to the pipeline.

Declaring a store per Region

```
"artifactStores": {
  "us-east-1": {
    "type": "S3", "location": "artifacts-111111111111-us-east-1",
    "encryptionKey": { "type": "KMS", "id": "alias/pipeline-use1" } },
  "eu-west-1": {
    "type": "S3", "location": "artifacts-111111111111-eu-west-1",
    "encryptionKey": { "type": "KMS", "id": "alias/pipeline-euw1" } }
}
```

5-7 A second bucket and a second key, both in eu-west-1.

4, 7 One key per Region: a key cannot be used outside its Region.

ON THE EXAM

Cross-account and cross-Region together: every Regional key policy must admit every target account's deploy role.

Four ways to reach many accounts

TOOL	BEST FOR	MODULE
CodePipeline cross-account actions	Releasing an application through ordered environments	This one
CloudFormation StackSets	One template into many accounts and Regions at once	Module 10
CDK Pipelines	CDK apps, with a pipeline that updates itself	Module 11
Control Tower customisations	Baselines for every account, by OU	Module 12

ON THE EXAM

An application moving dev, staging, production: a pipeline. The same baseline in every account: StackSets or Control Tower.

StackSets, in brief

- **Service-managed** StackSets use Organizations: no roles to create in each account.
- **Automatic deployment** to a target OU includes accounts added to it later.
- Maximum concurrent accounts, failure tolerance and Region order limit blast radius.
- A **delegated administrator** account can run them instead of the management account.

> ON THE EXAM
>
> 'Every account in this OU, including future ones, with least overhead': a service-managed StackSet with automatic deployment.

CDK Pipelines, in brief

- Each target account is **bootstrapped** with a trust to the tooling account.
- The pipeline is **self-mutating**: change its code, and it updates itself on the next run.
- **Stages** map to environments; **waves** deploy several stages in parallel.
- Deploying to an account that was never bootstrapped with that trust fails.

> ON THE EXAM
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> CDK pipeline cannot deploy into a new account: bootstrap it, trusting the tooling account.

Starting a pipeline from another account

- The source sits in account A; the pipeline in the tooling account.
- In account A, an **EventBridge rule** forwards the source event to the tooling account's event bus.
- The tooling bus's **resource policy** accepts events from account A.
- A rule on the tooling bus starts the pipeline, using a role allowed to do so.

> ON THE EXAM
>
> Pushes in another account do not start the pipeline: forward the event across accounts with EventBridge.

Releasing to many Regions in waves

01
First Region
Lowest traffic or least critical.

02
Bake
Alarms watch; a stage condition can roll back.

03
Wave two
A few Regions in parallel.

04
Bake again
Confidence grows with each wave.

05
The rest
Remaining Regions, still watched.

ON THE EXAM

A bad release should reach one Region, not all of them. Waves are how the pipeline limits blast radius.

Keeping cross-account power under control

- Deploy roles trust **only** the pipeline's service role, never the whole tooling account.
- A **separate deploy role** per environment, so a staging mistake cannot touch production.
- Manual approval before production accounts where change control requires it.
- SCPs can stop anyone but the pipeline from changing production resources.

> ON THE EXAM
>
> A trust policy naming the whole tooling account lets any principal there with AssumeRole deploy to production.

> A pipeline in a tooling account deploys a CloudFormation template to the production account. The action assumes a deploy role in production, which reads and decrypts the artifact successfully.
>
> The stack creation begins, but then fails: the stack events show access denied when creating a DynamoDB table.

What should the DevOps engineer fix?

A Add DynamoDB permissions to the pipeline's service role in the tooling account

B Add the production account to the artifact key's policy

C Add DynamoDB permissions to the CloudFormation execution role in the production account

D Add s3:GetObject for the deploy role to the artifact bucket policy

◆ PAUSE AND DECIDE

What should the DevOps engineer fix?

A **Pipeline service role**

It never creates resources in production. It only assumes the deploy role.

B **Key policy**

The scenario says decryption succeeded. That lock is already open.

C **CloudFormation execution role**

CloudFormation creates resources as this role. The stack started, so the failure is here.

D **Bucket policy**

The artifact was read successfully. That lock is already open too.

THE SIGNAL

Stack started, then a resource failed with access denied: the execution role. The scenario opened the other locks for you.

> A pipeline in us-east-1 deploys an application with CloudFormation. The team adds a second deploy action with its Region set to eu-west-1.
>
> Saving the pipeline fails with an error saying that no artifact store is configured for eu-west-1.

What should the DevOps engineer do?

A Enable S3 Cross-Region Replication from the us-east-1 artifact bucket to a bucket in eu-west-1

B Create a Multi-Region Access Point over the existing artifact bucket

C Create a second, separate pipeline in eu-west-1

D Add an artifact store for eu-west-1, with a bucket and a KMS key in that Region, to the pipeline

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A **Cross-Region Replication**

The pipeline does not use replicated buckets. It needs a store declared on itself.

B **Multi-Region Access Point**

A real S3 feature, but not something CodePipeline's artifact stores use.

C **Second pipeline**

Duplicates the whole release process and loses the single ordered flow.

D **Regional artifact store**

Exactly what the error asks for. CodePipeline then copies artifacts across automatically.

THE SIGNAL

Cross-Region action, no artifact store there: declare one on the pipeline. The copying is automatic.

> A company keeps its CodeCommit repository in a development account. The release pipeline lives in a central tooling account and uses that repository as its source through a cross-account role.
>
> The pipeline runs correctly when started by hand, but it never starts automatically when developers push to the main branch.

Which solution starts the pipeline on every push?

A. An EventBridge rule in the development account that forwards the repository's events to the tooling account's event bus, and a rule there that starts the pipeline

B. A Lambda function in the tooling account that polls the repository every minute and starts the pipeline

C. Move the pipeline into the development account

D. A notification rule on the repository that emails the release team

◆ PAUSE AND DECIDE

Which solution starts the pipeline on every push?

A Forward events across accounts

The push event fires in the development account. Forwarding it to the tooling bus lets a local rule start the pipeline.

B Polling Lambda

Works, with delay and custom code. Event forwarding is the native answer.

C Move the pipeline

Abandons the central tooling account the company chose.

D Email notification

Tells people; starts nothing.

THE SIGNAL

Source events fire in the wrong account: forward them with EventBridge. Allowed to send, willing to receive.

> A company uses AWS Organizations. It must deploy the same CloudFormation template, an IAM role and an AWS Config rule, to every account in its Workloads OU.
>
> Accounts added to that OU in future must receive the template automatically. The company does not use Control Tower and wants the least operational overhead.

Which solution meets these requirements?

A. A pipeline with one cross-account deploy action per account, updated as accounts are added

B. A service-managed StackSet targeting the Workloads OU, with automatic deployment enabled

C. A Lambda function triggered by account creation events that deploys the stack

D. A CDK pipeline with one stage per account

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A One action per account

Must be edited for every new account. Fails 'automatically'.

B Service-managed StackSet, auto-deploy

Targets the OU, deploys to future accounts automatically, needs no per-account roles.

C Lambda on account creation

Rebuilds what automatic deployment already does, and misses accounts moved into the OU.

D CDK stage per account

Every new account needs bootstrapping and a new stage. Not automatic.

THE SIGNAL

Same template, every account in an OU, including future ones: service-managed StackSet with automatic deployment.

Module 8, and Domain 1, in five lines

- Every boundary has locks on both sides: caller permissions, trust or resource policy, and the key.
- Cross-account artifacts need a customer managed key in the pipeline's Region.
- Action role starts the work; CloudFormation execution role creates resources.
- Cross-Region actions need a declared artifact store in each Region.
- Application releases: a pipeline. Same template everywhere: StackSets. Release Regions in waves.

> ON THE EXAM
>
> Domain 1 complete. Next, module 9 opens Domain 2 with CloudFormation for operators.